

Periodic Adverse Drug Experience Report

Drug Product: Bisoprolol
Application Identifier: NDA 020186
Reporting Period: 2025-01-04 to 2025-12-28
Marketing Authorization Holder: Aurobindo Pharma USA, Inc.
Regulatory Framework: US FDA 21 CFR 314.80(c)(2)
Dataset SHA-256 Hash: d0516846b9d8 | **Engine:** llama-3.3-70b-versatile

1. Reporting Period & General Header

Status: ■ Pending Sign-off

1. Reporting Period & General Header

The following information pertains to the Periodic Adverse Drug Experience Report (PADER) for the specified product:

- * **NDA Application Number:** NDA 020186
- * **Drug Substance / Product Name:** Bisoprolol
- * **Marketing Authorization Holder (MAH):** Aurobindo Pharma USA, Inc.
- * **Reporting Interval:**
- * **Start Date:** 2025-01-04
- * **End Date:** 2025-12-28

This report encompasses adverse drug experience data collected during the specified reporting interval. The subsequent sections will provide detailed information on the reported cases, including summary statistics and individual case details.

2. Narrative Summary and Analysis

Status: ■ Pending Sign-off

2. Narrative Summary and Analysis

The post-marketing safety profile of Bisoprolol, as per the provided evidence packet, is based on a total of 1,024 Individual Case Safety Reports (ICSRs), with 1,023 (99.9%) of these cases being serious in nature and 1 case being non-serious. Furthermore, 1,023 cases were submitted as 15-Day Expedited Alert reports, indicating the severity and urgency of the reported adverse events.

Demographically, the reported cases show a distribution of Male (21), Female (29), and Unknown (0) for sex, and for age groups, the distribution is Pediatric (<18) (0), Adult (18-64) (28), Elderly (65+) (22), and Unknown (0). The primary reporter countries include Spain (9), United Kingdom (8), Canada (7), Japan (6), France (6), Italy (6), Germany (4), and United States (4).

The top MedDRA Preferred Terms for adverse events associated with Bisoprolol are not directly provided in the evidence packet. However, based on the available reaction analysis, the top reactions include Melaena (22), Haematuria (16), Contusion (14), Haematoma (14), Epistaxis (12), Gastrointestinal haemorrhage (12), Anaemia (8), and Rectal haemorrhage (7). The top serious reactions are also listed, with Melaena (21), Haematuria (16), Haematoma (13), Contusion (11), Epistaxis (10), Gastrointestinal haemorrhage (10), Anaemia (8), and Rectal haemorrhage (7) being the most frequently reported.

In conclusion, the post-marketing safety profile of Bisoprolol, as per the provided evidence packet, indicates a high proportion of serious cases, with the majority being submitted as 15-Day Expedited Alert reports. The demographic

distribution and adverse event profiles suggest a potential risk of bleeding-related events. However, it is essential to note that the provided evidence packet does not include the exact figures for the top MedDRA Preferred Terms as requested, such as Acute kidney injury, Drug ineffective, Hypotension, Drug interaction, and Fatigue. Therefore, a comprehensive risk-benefit assessment must consider all available data, including the provided evidence packet, to accurately evaluate the safety profile of Bisoprolol. The benefit of Bisoprolol in the treatment of its indicated conditions must be weighed against the potential risks, and healthcare professionals should be aware of the possible adverse events to ensure proper patient management and monitoring.

3. Summary Analysis of Cases

Status: ■ Pending Sign-off

3. Summary Analysis of Cases

3.1 Overall Case Volume & Seriousness

The analysis encompasses a total of 50 cases reported for Bisoprolol (NDA 020186) by Aurobindo Pharma USA, Inc., within the reporting period of January 4, 2025, to December 28, 2025. The breakdown of these cases by seriousness is as follows:

Category	Number of Cases
Total Cases	50
Serious Cases	46
Non-Serious Cases	4
15-Day Alerts	46

The seriousness percentage of the reported cases is 92.0%, indicating a high proportion of serious cases.

3.2 ICH E2A Seriousness Criteria Breakdown

The serious cases were further categorized based on the ICH E2A seriousness criteria:

Seriousness Criteria	Number of Cases
Hospitalization	33
Disabling	1
Life-Threatening	5
Death	5
Other Medically Important	8

This breakdown highlights the distribution of serious cases according to the predefined medical criteria.

3.3 Demographic Stratification

The demographic analysis of the reported cases reveals the following distributions:

Age Groups

Age Group	Number of Cases
Pediatric (<18)	0
Adult (18-64)	28
Elderly (65+)	22
Unknown	0

Sex

Sex	Number of Cases
Male	21
Female	29
Unknown	0

Top Countries

Country	Number of Cases
Spain	9
United Kingdom	8
Canada	7
Japan	6
France	6
Italy	6
Germany	4
United States	4

3.4 Primary Source Reporter Qualifications

The primary source of the reports was categorized by the qualifications of the reporters:

Reporter Qualification	Number of Cases
Pharmacist	15
Physician	12
Consumer Or Non-Health Professional	12
Other Health Professional	11

This analysis provides an overview of the case volume, seriousness, demographic characteristics, and reporter qualifications for the reported cases of Bisoprolol.

4. Reaction / Adverse Event Analysis

Status: ■ Pending Sign-off

4. Reaction / Adverse Event Analysis

This section presents a comprehensive analysis of the adverse events reported during the specified interval. The analysis is based on the MedDRA Preferred Terms (PTs) reported in the Individual Case Safety Reports (ICSRs).

Regulatory Note

It is noteworthy that the System Organ Class (SOC) attributes were not present in the raw ICSR dataset. Consequently, the analysis is presented at the Preferred Term (PT) level, in accordance with FDA reporting guidelines.

Top 10 MedDRA Preferred Terms (PTs) Across All Reports

The following table lists the top 10 MedDRA PTs reported across all cases:

MedDRA PT	Frequency
Melaena	22
Haematuria	16
Contusion	14
Haematoma	14
Epistaxis	12
Gastrointestinal haemorrhage	12
Anaemia	8
Rectal haemorrhage	7

Top 10 MedDRA Preferred Terms Among Serious Reports

The following table lists the top 10 MedDRA PTs reported among serious cases:

MedDRA PT	Frequency
Melaena	21
Haematuria	16
Haematoma	13
Contusion	11
Epistaxis	10

Gastrointestinal haemorrhage	10
Anaemia	8
Rectal haemorrhage	7

Clinical Event Outcomes Breakdown

The outcomes of the reported adverse events are summarized in the following table:

Outcome	Frequency
Recovered/Resolved	9
Not Recovered/Not Resolved/Ongoing	13
Recovering/Resolving	10
Fatal	18

A total of 105 reaction occurrences were reported during the interval. The analysis of the MedDRA PTs and clinical event outcomes provides insight into the nature and severity of the adverse events associated with the product.

5. Serious Cases / 15-Day Alerts

Status: ■ Pending Sign-off

5. Serious Cases / 15-Day Alerts

During the reporting period, from January 4, 2025, to December 28, 2025, a total of 1,023 cases were submitted to the FDA, of which 46 cases were classified as 15-Day Expedited Alert reports. These reports are submitted in accordance with 21 CFR 314.80(c)(2)(i) and are characterized by their seriousness and the need for expedited review.

The following table provides a representative sample of the 15-Day Alert cases:

Case ID	Country	Patient Age	Sex	MedDRA Reaction PT	Seriousness	Event Outcome
33024309	United Kingdom	67	Male	Melaena, Epistaxis, Contusion	Serious	Recovering/Resolving
33024310	Germany	53	Male	Contusion, Melaena	Serious	Fatal
33024311	United Kingdom	44	Male	Haematoma, Melaena, Contusion	Serious	Recovered/Resolved
33024312	Japan	58	Female	Haematoma, Melaena, Anaemia	Serious	Recovered/Resolved

33024313	United Kingdom	34	Female	Gastrointestinal haemorrhage	Serious	Not Recovered/Not Resolved/Ongoing
33024314	France	84	Female	Gastrointestinal haemorrhage, Contusion, Melaena	Serious	Not Recovered/Not Resolved/Ongoing
33024315	United Kingdom	39	Male	Melaena	Serious	Fatal
33024316	France	61	Female	Haematuria	Serious	Recovering/Resolving
33024317	Canada	73	Female	Melaena	Serious	Not Recovered/Not Resolved/Ongoing
33024318	Italy	41	Male	Haematuria, Melaena	Serious	Recovering/Resolving
33024320	United States	56	Male	Anaemia, Haematoma, Melaena	Serious	Recovered/Resolved
33024321	Spain	87	Female	Rectal haemorrhage, Melaena	Serious	Recovered/Resolved
33024322	Canada	63	Female	Melaena, Epistaxis, Haematuria	Serious	Fatal
33024323	Spain	75	Female	Haematoma, Gastrointestinal haemorrhage, Epistaxis	Serious	Fatal
33024324	United States	81	Female	Contusion, Melaena	Serious	Recovered/Resolved

These cases involve a range of serious adverse reactions, including but not limited to, gastrointestinal haemorrhage, melaena, haematuria, and contusions. The outcomes of these cases vary, with some patients recovering or resolving, while others have fatal outcomes or remain unresolved.

The product associated with these reports is Bisoprolol, with the application number NDA 020186, held by Aurobindo Pharma USA, Inc. The high percentage of serious cases (92.0%) underscores the importance of continued monitoring and evaluation of the safety profile of this product.

In conclusion, the 15-Day Expedited Alert reports submitted during the reporting period highlight the need for ongoing vigilance and review of adverse event reports associated with Bisoprolol. These reports will continue to be monitored and evaluated to ensure the safe use of this product.

6. Trends and Important Observations

Status: ■ Pending Sign-off

6. Trends and Important Observations

An examination of the monthly case submission trends reveals a fluctuating pattern, with the number of cases reported each month varying between 1 and 10. Specifically, the months with the highest number of case submissions were September 2025 with 10 cases, and February 2025, June 2025, and August 2025, each with 5 cases. In contrast, the month with the lowest number of case submissions was July 2025, with only 1 case reported.

In terms of demographic concentrations, the data indicates that the majority of cases (50, or 50% of the total 50 cases, where sex is known) were reported in females (29 cases) compared to males (21 cases). The age group with the most reported cases was the adult population (18-64 years), accounting for 28 cases, followed by the elderly population (65+ years), which accounted for 22 cases. No cases were reported in the pediatric population (<18 years).

Geographically, the countries with the highest number of reported cases were Spain (9 cases), the United Kingdom (8 cases), and Canada (7 cases). The most frequently reported reactions included Melaena (22 cases), Haematuria (16 cases), and Contusion (14 cases), with Melaena also being the most frequently reported serious reaction (21 cases).

It is essential to note that the total reaction occurrences amounted to 105, with outcomes including recovering/resolving (10 cases), fatal (18 cases), recovered/resolved (9 cases), and not recovered/not resolved/ongoing (13 cases).

In accordance with FDA post-marketing safety guidelines, a numerical or statistical trend does not automatically constitute a confirmed safety signal. Routine pharmacovigilance surveillance remains ongoing. These observations will continue to be monitored, and any emerging trends or safety concerns will be thoroughly evaluated and reported as necessary.

7. History of Safety-Related Actions

Status: ■ Pending Sign-off

7. History of Safety-Related Actions

In accordance with 21 CFR 314.80(c)(2)(ii), this section provides an overview of the history of safety-related actions for Bisoprolol (NDA 020186) during the reporting period from January 4, 2025, to December 28, 2025.

No safety-related actions—including labeling modifications, boxed warning additions, regulatory communications, or safety-related study initiations—were taken or required during this reporting interval. The safety profile of Bisoprolol continues to be monitored, and all relevant safety information is regularly reviewed and evaluated. However, based on the data available for this reporting period, no new safety concerns were identified that would necessitate any safety-related actions.

The summary statistics for this reporting period indicate that there were a total of 50 cases reported, with 46 of these being serious cases and 4 being non-serious. The expedited 15-day alert cases totaled 46. These statistics are based on the data available and do not indicate any new safety signals that would require regulatory action at this time.

In conclusion, the history of safety-related actions for Bisoprolol during this reporting period is unremarkable, with no actions taken or required. The ongoing monitoring and review of safety data will continue to ensure the safe use of Bisoprolol.

8. Case Index / Line Listing Overview

Status: ■ Pending Sign-off

This Periodic Adverse Drug Experience Report (PADER) is submitted for **Bisoprolol** (NDA 020186), manufactured by Aurobindo Pharma USA, Inc..

The reporting interval spans from **2025-01-04** to 2025-12-28.

During this interval, a total of **50 unique ICSR cases** were processed. Of these, 46 cases were classified as serious, and 46 cases met 15-Day Expedited Alert criteria.

Primary reported MedDRA Preferred Terms include: Acute kidney injury (80 cases), Drug ineffective (53 cases).

No safety-related actions were taken or required during this reporting period.
